

Data Science in Cybersecurity Final Project

Required PDF Report - Fraud Detection Handbook reproducibility and critical evaluation

GitHub branch: <https://github.com/PolarScorpion/data-science-in-cyber/tree/cyber-final-project>

Key files: README.md; notebooks/main_analysis.ipynb; reports/final_report.md; reports/final_synthesis.md; reports/behavioral_features.md; reports/baseline_modeling.md; reports/eda_findings.md; reports/reproducibility_audit.md

1. Summary

This project reproduces and critically evaluates a fraud-detection workflow from the Fraud Detection Handbook using its public simulated transaction dataset. The scope is limited to simulated transaction data, baseline modeling, temporal validation, fraud-appropriate metrics, and behavioral feature engineering.

The central research question is whether the handbook's conclusions about temporal validation, metric choice, and behavioral features remain supported under the simulated fraud scenarios, and how far those findings can generalize beyond data generated by known rules.

Project workflow summary

Phase	Output
1	Repository scaffold and ignore rules
2	Source selection and reproducibility audit
3	Data loading and initial inspection
4	Exploratory data analysis
5	Baseline fraud modeling
6	Leakage-safe behavioral feature engineering
7	Final synthesis and external-validity analysis
8	Submission-readiness audit and cleanup

2. Critical Evaluation

The strongest evidence is internal reproducibility on a public synthetic benchmark, not proof of production fraud performance. The train, validation, and test periods are chronological, which prevents random temporal leakage and better matches the online scoring setting.

- Fraud prevalence is below 1%, so AP / PR-AUC is the primary metric.
- Accuracy is misleading and is not used as the main metric.
- Thresholds are selected on validation only, and final metrics are evaluated on the held-out test period only.

- Synthetic data may overstate deployability because engineered features can align with the simulator rules that generated the labels.
- The project does not claim production readiness or real-world fraud performance.

3. Feature Engineering Analysis

The baseline feature set uses transaction amount, log amount, and cyclic encodings of hour and day. The past-only behavioral feature set adds customer and terminal histories computed only from earlier transactions, such as prior counts, prior amount summaries, recent activity windows, time since previous transaction, and customer-terminal interaction counts.

The prior fraud-label history feature set adds previous fraud counts and fraud rates for customers and terminals. These features give much larger gains, but they require timely known prior labels. If labels arrive after review, disputes, or chargeback windows, those features would need explicit lagging or exclusion.

Leakage controls include sorting by transaction time and transaction id, computing histories from past transactions only, excluding TX_FRAUD and TX_FRAUD_SCENARIO as predictors, and avoiding full-window customer or terminal aggregates.

4. Reproducibility Analysis

The source code, notebooks, and simulated dataset instructions are public. Raw data are obtained separately from the Fraud Detection Handbook dataset repository and are not stored in this submission repository. The notebook and source files reproduce data loading, exploratory analysis, baseline modeling, behavioral features, and saved metric tables without requiring fitted model binaries.

- Dataset source: <https://github.com/Fraud-Detection-Handbook/simulated-data-raw>
- Original handbook: <https://fraud-detection-handbook.github.io/fraud-detection-handbook/>
- Original code repository: <https://github.com/Fraud-Detection-Handbook/fraud-detection-handbook>
- Submission branch: <https://github.com/PolarScorpion/data-science-in-cyber/tree/cyber-final-project>

The main remaining reproducibility limitation is external validity: the data are synthetic and generated from known fraud scenarios, so high performance may partly reflect recovery of simulator structure rather than robust real-world fraud behavior.

5. Experimental Results

The final test period contains 345,144 transactions and 3,091 fraud cases, for 0.896% fraud prevalence. Average Precision / PR-AUC is emphasized because this is an alert-ranking problem under extreme imbalance. ROC-AUC, precision, recall, F1, confusion matrix counts, and Precision@k are also reported in the project reports.

Feature set	Best model	AP / PR-AUC	ROC-AUC	Precision	Recall	F1
Baseline transaction/time features	Logistic regression	0.245	0.644	0.940	0.211	0.345
Past-only behavioral features, no prior fraud labels	Histogram gradient boosting	0.257	0.658	0.561	0.289	0.381
Past behavioral features + prior fraud-label history	Histogram gradient boosting	0.870	0.988	0.890	0.774	0.828

Past-only behavioral features improve performance modestly. Prior fraud-label history gives much larger gains, but that result depends on the operational assumption that earlier fraud labels are known before later transactions are scored.

MCC is mentioned in the assignment's recommended metrics but was omitted from the implemented evaluation. The project prioritized AP / PR-AUC, ROC-AUC, precision, recall, F1, confusion matrix counts, and Precision@k because the task is alert ranking under extreme imbalance. MCC can be added as a minor extension.

6. Error Analysis

Error analysis is based on held-out chronological test-set confusion counts and alert-ranking behavior saved in the project result tables. These examples interpret the observed errors; they are not new model runs or additional experiments.

Feature set	Best model	AP / PR-AUC	Precision	Recall	F1	FP	FN	TP	Alerts
Baseline transaction/time features	Logistic regression	0.245	0.940	0.211	0.345	42	2,438	653	695
Past-only behavioral features, no prior fraud labels	Histogram gradient boosting	0.257	0.561	0.289	0.381	699	2,198	893	1,592
Past behavioral features + prior fraud-label history	Histogram gradient boosting	0.870	0.890	0.774	0.828	295	699	2,392	2,687

False positives are legitimate transactions flagged as suspicious. They create review cost, customer friction, and possible unnecessary declined or blocked activity. A likely false-positive pattern is a legitimate transaction that looks unusual in amount, timing, terminal behavior, or customer history.

False negatives are fraudulent transactions missed by the detector. They imply direct fraud loss, delayed intervention, and continued compromised customer or terminal activity. A likely false-negative pattern is fraud that resembles normal customer behavior or appears before enough behavioral history has accumulated.

The baseline transaction/time model is conservative: AP / PR-AUC 0.245, precision 0.940, recall 0.211, and F1 0.345, with 42 false positives and 2,438 false negatives. Past-only behavioral features improve AP / PR-AUC to 0.257, recall to 0.289, and F1 to 0.381, but precision falls to 0.561 and false positives rise to 699. Prior fraud-label history has the strongest test result, AP / PR-AUC 0.870, precision 0.890, recall 0.774, and F1 0.828, but depends on timely known prior labels.

Prior-label features have specific possible failure modes. A first fraud in a new pattern may occur before previous labels are known, producing false negatives. Previous fraud history can also make later legitimate behavior look suspicious, producing false positives. These are plausible operational patterns consistent with the feature design, not proven real-world failure modes.

The FP/FN trade-off is central: higher recall usually increases review burden, while higher precision may miss more fraud. The best threshold depends on business and security costs, not only F1. This project uses validation-set F1 thresholding as a reproducible benchmark, not as a final deployment policy.

7. Executive Summary

- Best conservative result: past-only behavioral features modestly improve the baseline.
- Best numerical result: prior fraud-label history reaches much higher AP / PR-AUC but requires timely known labels.
- Main methodological contribution: leakage-aware temporal evaluation with fraud-appropriate metrics.
- Main limitation: synthetic data may make the feature-label relationships cleaner than real payment data.

8. Summing It Up

This submission is best understood as a reproducible and critical benchmark study. It shows why temporal validation, precision-recall evaluation, and past-only feature construction matter in fraud detection. The strongest model results are internally impressive but externally conditional, especially when they depend on prior fraud labels that may not be available at authorization time.

The project supports three limited conclusions. First, chronological validation is necessary for fraud data with time-dependent behavior. Second, precision-recall metrics are more informative than accuracy when fraud prevalence is below 1%. Third, past behavioral histories are useful, but the interpretation changes sharply when features depend on prior known labels.

The project does not prove real-world deployability. Delayed labels, calibration, drift, alert-budgeting, review cost, privacy controls, and cost-sensitive thresholds remain future work.